PM2.5 Exposure and Per Capita Health Expenditure have a significant relationship with Infant Mortality Rates in South and Southeast Asia

Pollution, Policy, and Health: Exploring Infant Mortality Trends in South and Southeast Asia

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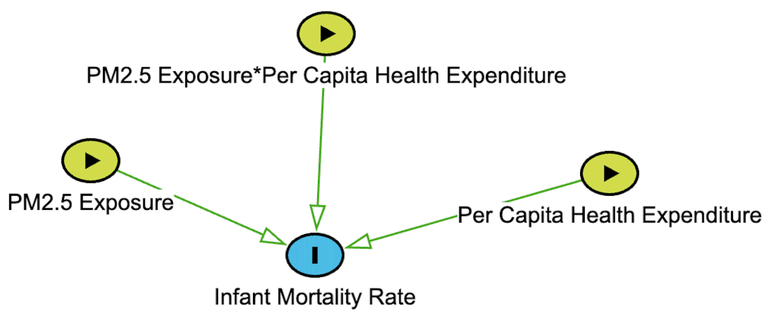
Introduction

Air pollution has become a major global health threat, with a 2018 WHO report showing that 9 out of 10 people worldwide breathe polluted air, causing 7 million deaths annually. Fine particulate matter (PM2.5), with a diameter less than 2.5 microns, is considered the most hazardous pollutant, linked to 4.2 million deaths and 103 million disability-adjusted life-years globally in 2015 [1].

PM2.5 exposure is also associated with rising infant mortality rates, particularly affecting children under five due to their developing respiratory systems and mouth-breathing [2]. These effects are more severe in developing regions like Asia, where 92% of the population is exposed to dangerous air pollution levels [3].

South and Southeast Asia face acute pollution issues driven by rapid industrialization, urbanization, population growth, and reliance on coal and solid fuels. Frequent smog and haze episodes result from high PM2.5 concentrations caused by combustion emissions [4]. Moreover, limited public health spending in these regions, often due to their recent growth and governance models, exacerbates infant mortality, despite evidence linking higher per capita health spending to lower IMR in Asia [^4].

Model



To identify the causal effect of PM2.5 exposure and per-capita health expenditure on infant mortality rates, I include PM2.5 exposure, per-capita health expenditure, and their interaction term in the model as explanatory variables. The interaction term allows me to assess whether health expenditure mitigates/moderates the effect of PM2.5 exposure in countries where the latter is high. To identify and compare the effects of PM2.5 exposure and health expenditure for South and Southeast Asia, I use a similar regression for each region.

$$inf.mortality = \beta_0 + \beta_1(pm2.5) + \beta_2(pcap.health.exp) + \beta_3(pm2.5 * pcap.health.exp) + \epsilon_i$$

Data

For the purpose of this research, I used data from the World Development Indicators (WDI) website, which is the World Bank's primary collection of development indicators, compiled from recognized international sources. I extracted data about 16 countries in total – 7 from South Asia (Afghanistan, Bangladesh, Bhutan, India, Nepal, Pakistan, Sri Lanka) and 9 from Southeast Asia (Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, the Philippines, Singapore, Thailand, Vietnam) – between the years 2000 and 2020. The data contains information about the following variables: country, year, infant_mortality(infant mortality rate per 1000 live births), pm2.5 (mean annual exposure to pm2.5), health_exp_pcap (per capita government health expenditure).

Results

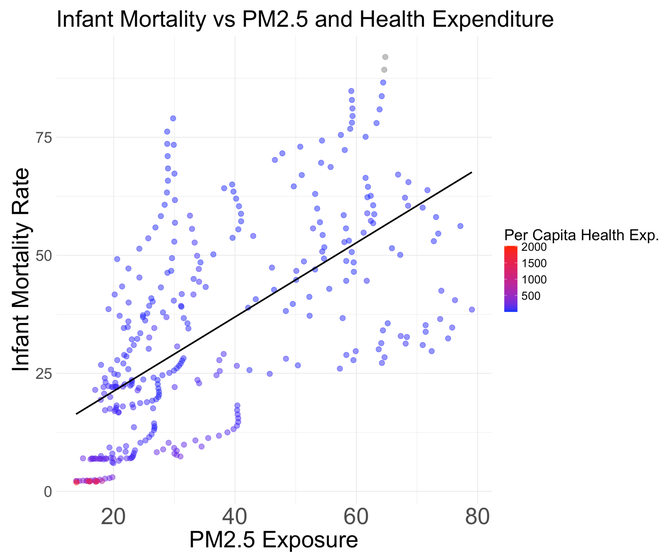


Figure 1: Infant Mortality vs PM2.5 and Health Expenditure in S. and SE Asia

We can see that countries with higher levels of health expenditure (red) tend to cluster at lower infant mortality rates, while regions with low per capita health expenditure seem to have higher IMR even at low to moderate PM2.5 exposure levels.

Table 1: Regression Results for Infant Mortality in South Asia

term	estimate	std.error	statistic	p.value
(Intercept)	34.1572	8.4536	4.0406	0.0001
pm2.5	0.3606	0.1479	2.4384	0.0160
health_exp_pcap	-0.2806	0.2191	-1.2804	0.2025
pm2.5:health_exp_pcap	-0.0054	0.0062	-0.8759	0.3826

The estimate of the interaction term pm2.5*health_exp_pcap is -0.0054, which suggests that the effect of PM2.5 exposure on infant mortality decreases slightly when per capita health expenditure increases.H However, the interaction effect lacks statistical significance (p-value = 0.3826), suggesting that there is insufficient evidence for this conclusion. The R-squared of this regression model was 0.4701, implying moderate explanatory power.

Table 2: Regression Results for Infant Mortality in Southeast Asia

term	estimate	std.error	statistic	p.value
(Intercept)	-6.0315	3.8280	-1.5756	0.1173
pm2.5	1.4535	0.1405	10.3440	<1e-04
health_exp_pcap	0.1612	0.0188	8.5754	<1e-04
pm2.5:health_exp_pcap	-0.0112	0.0012	-9.7222	<1e-04

This regression output table suggests that the interaction term is highly significant. For every 1-unit increase in per capita health expenditure, the positive effect of PM2.5 exposure on infant mortality reduces by 0.0112 per 1000 live births. This means that health expenditure has a strong moderating effect on infant mortality in the presence of PM2.5 exposure in Southeast Asia, i.e. healthcare plays a protective role in regions with high pollution. The overall model fit is better for Southeast Asia than for Southeast Asia, with an R-squared of 0.637.

Table 3: Welch Two Sample t-test Results

Statistic	Value
t-statistic	-5.3055
Degrees of Freedom	147.66
p-value	4.034e-07
95% Confidence Interval	[-183.68, -83.98]
Mean of South Asia	18.83803
Mean of Southeast Asia	152.667

There are also differences between the per capita health expenditure in South Asia and Southeast Asia. The difference between the estimates of per capita health expenditures is also statistically significant, as shown by the results of the t-test. The p-value of the t-test is less than 0.05, allowing us conclude that there is a significant difference in the healthcare expenditures of the two regions. Moreover, the negative confidence interval and negative t-statistic further solidifies that per capita healthcare expenditure in Southeast Asia is significantly higher compared to South Asia.

Discussion

The objective of my analysis was to investigate the relationship between PM2.5 exposure, healthcare expenditure, and infant mortality rates (IMR) in South and Southeast Asia. More specifically, we aimed to investigate whether healthcare expenditure could moderate the effect of high exposure of PM2.5 on IMR. Ultimately, the results of my analysis provide partial support for the theoretical framework that both PM2.5 exposure and healthcare expenditure influence infant mortality, but with some region-specific dynamics between these variables.

The significant relationship between the interaction of PM2.5 exposure and per capita health expenditure, and IMR in Southeast Asia suggests that higher levels of health expenditure are associated with a weaker effect of PM2.5 exposure on IMR. This implies that regions with more robust healthcare infrastructure and greater financial resources may be able to provide better-quality healthcare that mitigates the effect of high pollution levels. In contrast, the lack of significance for the interaction term in South Asia suggests a more unclear relationship in this region. The significantly lower health expenditure in South Asia, instead of mitigating, exacerbates the effect of PM2.5 exposure on IMR. Moreover, this could also be due to other unobserved factors such as quality of life, per capita income, poverty rates, etc. Therefore, the effects of healthcare expenditure may be outweighed by structural challenges in South Asia.

A major limitation of this study is that it fails to capture the individual variation between countries in these regions. For instance, the per capita health expenditure in certain South Asian countries, such as Sri Lanka and Bhutan, is much higher as compared to neighboring countries, such as Bangladesh and Nepal. Additionally, Bangladesh and India are more densely populated than Sri Lanka and Bhutan, which could affect the allocation and effectiveness of health expenditure levels. This study also fails to account for potential confounders such as access to clean technologies, quality of healthcare, or implementation of air pollution control policies, which may significantly affect IMR. This study, however, opens up avenues for future research that focuses on other factors that affect the relationship between pollution and infant mortality rate.

References

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